

Problem Set 4
COMP301 Spring 2025
Week 5: 10.3.2025 - 17.3.2025

Instructions:

- Submit your answers to the LearnHub PS4 assignment until **March 15th** Saturday, at 23.59.
- Please submit only **one single PDF file**, where all of your codes for each of the parts are included.
- Name your submission file as *id_username_ps4.pdf*
(Example: *00000_bcosar18_ps4.pdf*).

Review Part.

Problem 1: Define the set of square numbers (0, 1, 4, 9, 16 ...) by using top-down, bottom-up, and rules of inference approach. You can check if an element of the set can be decomposed into an equation that satisfies the properties. For equations, you may need to use an additional variable and demonstrate the relation between the variables. In your answer do not use the square operator. However, you are allowed to use square root. Hint: $(x + y)^2 = x^2 + 2xy + y^2$

Problem 2:

a) A binary tree with numeric leaves and interior nodes labeled with symbols may be represented using three-element lists for the interior nodes by the grammar:

Definition 1.1.7 (binary tree)

$\text{Bintree} ::= \text{Int} \mid (\text{Symbol Bintree Bintree})$

Here are some examples of such trees:

1
(foo 1 2)

In this part, you need to write down 3 valid Binary Tree examples considering above definition.

b) The *lambda calculus* is a simple language that is often used to study the theory of programming languages. This language consists only of variable references, procedures that take a single argument, and procedure calls. We can define it with the grammar:

Definition 1.1.8 (lambda expression)

$\text{LcExp} ::= \text{Identifier}$
 $\quad ::= (\text{lambda } (\text{Identifier}) \text{ LcExp})$
 $\quad ::= (\text{LcExp LcExp})$

where an identifier is any symbol other than `lambda`.

In this part, you need to write down 3 valid lambda expression examples considering above definition.

Coding Part.

Problem 3: Given a nested list and an input, implement a procedure named "count-occurrence-nested" that counts the occurrence of the given element in the nested list.

```
(count-occurrence-nested '(a b (a b (a b)) (a b)) 'a) ; returns 4  
(count-occurrence-nested '(a (b a) (a b) (a b c)) 'b) ; returns 3
```

- Problem 1:
 - A is a set of square numbers iff:
 - 0 is in A
 - For any element e in A, $[e+1+2\sqrt{e}]$ is also in A.
 - The set of square numbers is the smallest set that satisfies
 - 0 is in the set of square numbers;
 - For any element e in the set of square numbers, $[e+1+2\sqrt{e}]$ is also in the set of square numbers

$$\frac{\boxed{}}{0 \in \text{Set}} \quad \frac{e \in \text{SetSN}}{[e+1+2\sqrt{e}] \in \text{SetSN}}$$

- Problem 2:

(A)

- (foo (bar 1 2) 3)
- (bar (foo 2 (foo 3 1)) (bar 1 2))
- (foo (foo (foo (foo 1 2) 3) 4) 5)

(B)

$\text{LcExp} ::= \text{Identifier}$

$\quad ::= (\text{lambda } (\text{Identifier}) \text{ LcExp})$

$\quad ::= (\text{LcExp LcExp})$

-
- x
- (lambda (x) (square x)) => (square x) can be decomposed as (identifier identifier)
- (foo (lambda (y) (sqrt (bar y)))) => (Identifier (lambda (identifier) (identifier (identifier identifier))))

- Problem 3:

```
(define (count-occurrencenested list x) (cond
  ((null? list) 0)
  ((list? (car list)) (+
    (count-occurrencenested (car list) x)
    (count-occurrencenested (cdr list) x)
  ))
  (else (if (equal? x (car list))
    (+ 1 (count-occurrencenested (cdr list) x))
    (count-occurrencenested (cdr list) x)))))
```